Algebra 1
Unit 10
Lesson 1 Homework

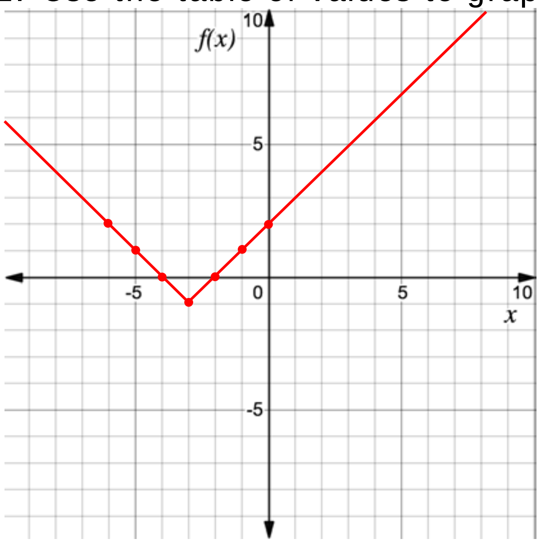
Name____________________
Date____________________
Period__________________

For questions 1-8, consider the absolute value function $f(x) = |x + 3| - 1$.

1. Complete the table of values for $f(x) = |x + 3| - 1$.

x	-6	-5	-4	-3	-2	-1	0
$f(x)$	2	1	0	-1	0	1	2

2. Use the table of values to graph $f(x) = |x + 3| - 1$ on the coordinate grid.



3. What is the vertex of $f(x) = |x + 3| - 1$?
   $(-3, -1)$

4. What is the equation for the axis of symmetry of $f(x) = |x + 3| - 1$?
   $x = -3$

5. What are the x and y-intercept(s) of $f(x) = |x + 3| - 1$?
   x-intecept: $(-4,0)$ and $(-2,0)$
   y-intercept: $(0,2)$

6. What are the domain and range of $f(x) = |x + 3| - 1$?
   D: R
   R: $y \geq -1$

7. Identify the intervals on which the function $f(x) = |x + 3| - 1$ is increasing and/or decreasing.
   increasing: $x > -3$ or $(-3, \infty)$
   decreasing: $x < -3$ or $(-\infty, -3)$

8. Identify the intervals on which the function $f(x) = |x+3| - 1$ is positive and negative.

Positive $(-\infty, -4) \cup (-2, \infty)$

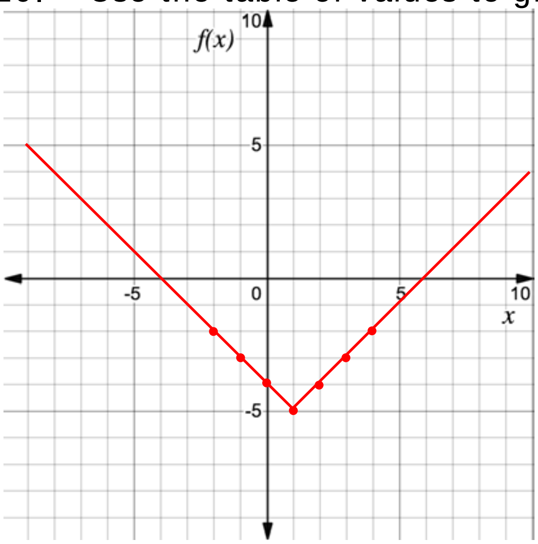
Negative $(-4, -2)$

For questions 9-10, consider the absolute value function $f(x) = |x-1| - 5$.

9. Complete the table of values for $f(x) = |x-1| - 5$.

x	-2	-1	0	1	2	3	4
$f(x)$	-2	-3	-4	-5	-4	-3	-2

10. Use the table of values to graph $f(x) = |x-1| - 5$ on the coordinate grid.



Algebra 1 Name____________________
Unit 10 Date____________________
Lesson 2 Homework Period____________________

Use the following context to answer questions 1-5.

The average mass of a baseball is $8\frac{7}{8}$ ounces (oz) with a tolerance of $3\frac{7}{8}$ oz. If x is the mass of the baseball and V is the variation from $8\frac{7}{8}$ oz., the function $V(x) = \left|x - 8\frac{7}{8}\right|$ can be used to find the amount of variation.

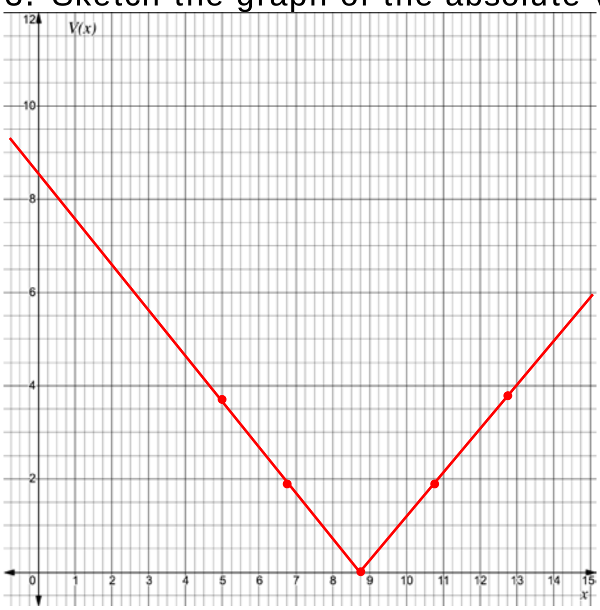
1. Determine the maximum and minimum x-values.

   minimum: $8\frac{7}{8} - 3\frac{7}{8} = 5$ maximum: $8\frac{7}{8} + 3\frac{7}{8} = 12\frac{3}{4}$

2. Complete the table for $V(x) = \left|x - 8\frac{7}{8}\right|$. Use the minimum x-value in the first column, the maximum x-value in the last column, and add $\frac{31}{16}$ to each x-value in between.

x	5	$6\frac{15}{16}$	$8\frac{7}{8}$	$10\frac{13}{16}$	$12\frac{3}{4}$
y	$3\frac{7}{8}$	$1\frac{15}{16}$	0	$1\frac{15}{16}$	$3\frac{7}{8}$

3. Sketch the graph of the absolute value function



4. Interpret the vertex of $V(x) = \left|x - 8\frac{7}{8}\right|$.

   The average mass of a baseball.

5. Interpret the axis of symmetry of $V(x) = \left|x - 8\frac{7}{8}\right|$.

   The axis of symmetry at $x = 8\frac{7}{8}$ helps to determine how far from average a baseball is.

Use the following context to answer questions 6-10.

The clearance below a bridge is 190.5 feet (ft) above sea level. According to the distribution of tidal phases, the water under the bridge can fluctuate up to 4.5 feet above or below the normal depth. If x is the amount of clearance the bridge has above the water, in ft, and W is the variation in water depth away from sea level, in ft, the equation to find the amount of variation is $V(x) = |x - 190.5|$.

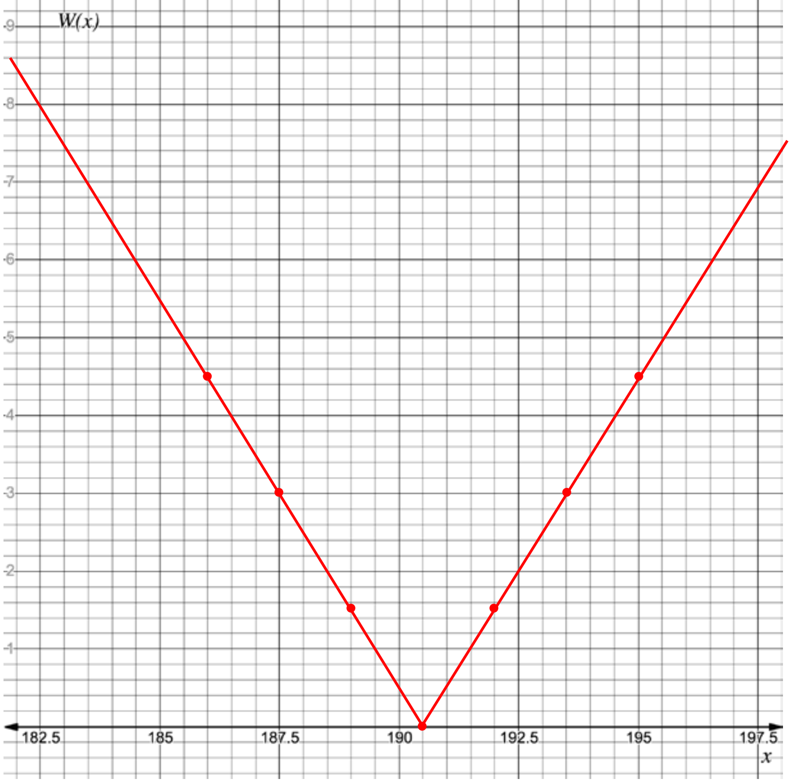
6. Determine the maximum and minimum clearance between the bridge and water.

   Minimum: $190.5 - 4.5 = 186$ Maximum: $190.5 + 4.5 = 195$

7. Complete the table for $V(x) = |x - 190.5|$.

x	186	187.5	189	190.5	192	193.5	195
y	4.5	3	1.5	0	1.5	3	4.5

8. Sketch the graph of the absolute value function



9. Interpret the vertex of $V(x) = |x - 190.5|$.

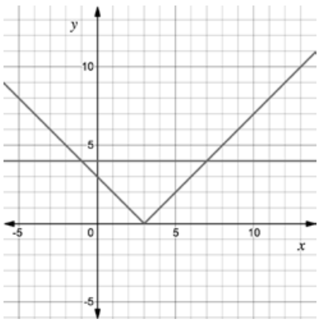
   The clearance below a bridge with no variations in sea level.

10. Interpret the axis of symmetry of $V(x) = |x - 190.5|$.

    Helps to determine the fluctuation in sea level when it is not the norm at $x = 190.5$.

Algebra 1 Name____________________
Unit 10 Date____________________
Lesson 3 Homework Period____________________

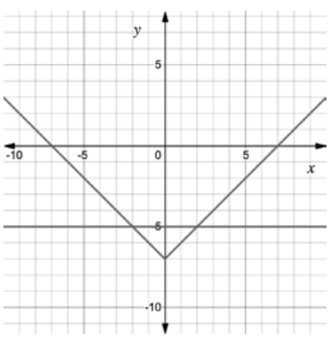
For questions 1-2, consider the absolute value equation $|x-3|=4$. The graphs of $y=|x-3|$ and $y=4$ are shown on the coordinate grid.



1. Use the graph to determine the solution to $|x-3|=4$.
   $x=-1$ and $x=7$y

2. Solve $|x-3|=4$ algebraically.
   $x-3=4$ $\quad$ $x-3=-4$
   $+3 \quad +3$ $\quad$ $+3 \quad +3$
   $x=7$ $\quad$ $x=-1$

For questions 3-4, consider the absolute value equation $|x|-7=-5$. The graphs of $y=|x|-7$ and $y=-5$ are shown on the coordinate grid.



3. Use the graph to determine the solution to $|x|-7=-5$.
   $x=-2$ and $x=2$

4. Solve $|x|-7=-5$ algebraically.
   $|x|-7=-5$ $\quad$ $|x|=2$
   $+7 \quad +7$ $\quad$ $x=2 \quad x=-2$

For questions 5-10, solve each absolute value equation.

5. $|m| - 29 = -17$

$+29 \quad + 29$

$|m| = 12$

$m = 12 \quad m = -12$

6. $|h - 16| = 21$

$h - 16 = 21 \qquad h - 16 = -21$

$+16 \; + 16 \qquad +16 \quad + 16$

$h = 37 \qquad h = -5$

7. $|6x| = 21$

$\frac{6x}{6} = \frac{21}{6} \qquad \frac{6x}{6} = \frac{-21}{6}$

$x = 3.5 \qquad x = -3.5$

8. $11 = |x| + 8$

$-8 \qquad -8$

$|x| = 3$

$x = 3 \qquad x = -3$

9. $|k - 51| = 72$

$k - 51 = 72 \qquad k - 51 = -72$

$+51 \quad +51 \qquad +51 \quad + 51$

$k = 123 \qquad k = -21$

10. $20 = |8x|$

$\frac{8x}{8} = \frac{20}{8} \qquad \frac{8x}{8} = \frac{-20}{8}$

$x = 2.5 \qquad x = -2.5$

Algebra 1 Name____________________
Unit 10 Date____________________
Lesson 4 Homework Period____________________

Use the following context to answer questions 1-5.

The length of a machine part should be 5.25 inches (in). The manufacturing tolerance is within 0.03 in.

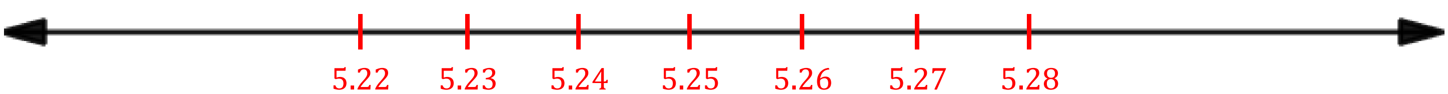
1. What is the minimum value the machine part could be?

$5.25 - 0.03 = 5.22$

2. What is the maximum value the machine part could be?

$5.25 + 0.03 = 5.28$

3. Sketch a number line that shows the minimum value, maximum value, and 5.25.



4. Write an absolute value equation that could be used to determine the maximum and minimum value of m, the length of the machine part, algebraically.

$$|m - 5.25| = 0.03$$

5. Show that the solutions to the equation match the maximum and minimum values determined in questions 1 and 2.

$$\begin{aligned} m - 5.25 &= 0.03 \\ +5.25 \quad &+5.25 \\ m &= 5.28 \end{aligned}$$

$$\begin{aligned} m - 5.25 &= -0.03 \\ +5.25 \quad &+5.25 \\ m &= 5.22 \end{aligned}$$

For questions 6-10, solve each absolute value equation.

6. $0 = 3|x+2| - 15$

$\frac{3|x+2|}{3} = \frac{15}{3}$

$|x+2| = 5$

$x + 2 = 5 \qquad x + 2 = -5$

$\quad -2 \ -2 \qquad\quad -2 \ -2$

$x = 3 \qquad x = -7$

7. $|-4x+2| - 9 = 0$

$\qquad\qquad +9 \ \ +9$

$|-4x+2| = 9$

$-4x + 2 = 9 \qquad -4x + 2 = -9$

$\qquad -2 \ -2 \qquad\qquad -2 \ -2$

$\frac{-4x}{-4} = \frac{7}{-4} \qquad \frac{-4x}{-4} = \frac{-11}{-4}$

$x = -1.75 \qquad x = 2.75$

8. $|x+5| + 9 = 3$

$\qquad\quad -9 \ \ -9$

$|x+5| = -6$

No Solution

9. $|3x+4| - 6 = 25$

$\qquad\qquad +6 \ \ +6$

$|3x+4| = 31$

$3x + 4 = 31 \qquad 3x + 4 = -31$

$\qquad -4 \ -4 \qquad\qquad -4 \ -4$

$\frac{3x}{3} = \frac{27}{3} \qquad \frac{3x}{3} = \frac{-35}{3}$

$x = 9 \qquad x = -11\frac{2}{3}$

10. $-|x-3| + 7 = 0$

$\qquad\qquad -7 \ -7$

$\frac{-|x-3|}{-1} = \frac{-7}{-1}$

$|x-3| = 7$

$x - 3 = 7 \qquad x - 3 = -7$

$\quad +3 \ +3 \qquad\quad +3 \ +3$

$x = 10 \qquad x = -4$

Algebra 1 Name Answer Key ____________
Unit 10 Date____________________
Lesson H1 Homework Period___________________

For questions 1-2, consider the absolute value inequality $|4x-9|-8<19$.

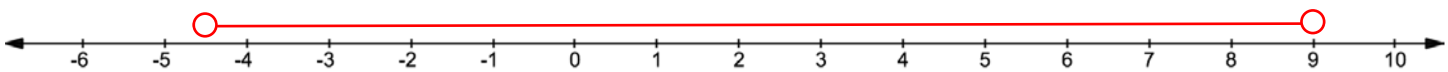
1. Solve the inequality.

$|4x-9|-8<19$
$\quad +8 \quad +8$
$|4x-9|<27$
$-4.5<x<9$

$4x-9<27$
$\quad +9 \quad +9$
$4x<36$
$x<9$

$4x-9>-27$
$\quad +9 \quad +9$
$4x>-18$
$x>-4.5$

2. Graph the solution on the number line.



For questions 3-4, consider the absolute value inequality $2|x-5|-4<0$.

3. Solve the inequality.

$\frac{2|x-5|}{2}<\frac{4}{2}$

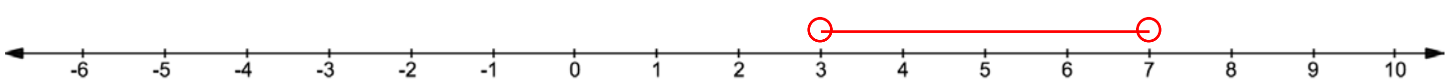
$|x-5|<2$

$x-5<2$
$\quad +5 \quad +5$
$x<7$

$x-5>-2$
$\quad +5 \quad +5$
$x>3$

$3<x<7$

4. Graph the solution on the number line.



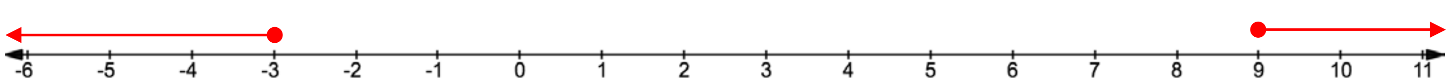
For questions 5-6, consider the absolute value inequality $|3x-9|\geq 18$.

5. Solve the inequality.

$3x-9\geq 18$
$\quad +9 \quad +9$
$3x\geq 27$
$x\geq 9$

$3x-9\leq -18$
$\quad +9 \quad +9$
$3x\leq -9$
$x\leq -3$

6. Graph the solution on the number line.



For questions 7-10, solve the absolute value inequality.

7. $-8|x-7| > 32$

$\overline{-8} \quad \overline{-8}$

$|x-7| < -4$

No solution

8. $-3|x-5| - 6 \leq 9$

$+6 \quad +6$

$\frac{-3|x-5|}{-3} \leq \frac{15}{-3}$

$|x-5| \geq -5$

Infinitely many solutions

9. $|x-11| \geq -3$

Infinitely many solutions

10. $|7x+4| - 6 < 8$

$+6 \quad +6$

$|7x+4| < 14$

$7x + 4 < 14$

$-4 \quad -4$

$\frac{7x}{7} < \frac{10}{7}$

$x < 1\frac{3}{7}$

$7x + 4 > -14$

$-4 \quad -4$

$\frac{7x}{7} > \frac{-18}{7}$

$x > -2\frac{4}{7}$

Algebra 1 Name Answer Key____________
Unit 10 Date___________________
Lesson H2 Homework Period__________________

Use the following context to answer questions 1-4.

Members of the cross-country team are preparing for a 3-mile race. The average time for the team is 20.5 minutes. The fastest and slowest times varied from the average by 3.4 minutes.

1. Define the variable.

   Let m represent the minutes ran.

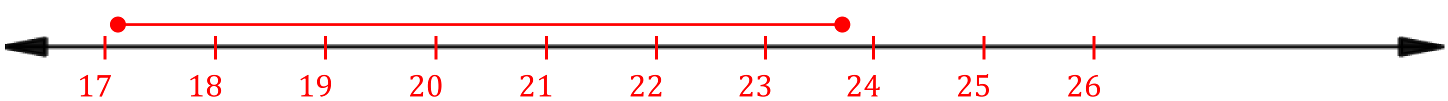
2. Write an absolute value inequality that represents the context.

   $|m - 20.5| \leq 3.4$

3. Solve the inequality. Write the answer as a compound inequality.

   $m - 20.5 \leq 3.4$ $\quad$ $m - 20.5 \geq -3.4$
   $\quad +20.5 \quad +20.5$ $\quad$ $+20.5 \quad +20.5$
   $m \leq 23.9$ $\quad$ $m \geq 17.1$ $\quad$ $17.1 \leq m \leq 23.9$

4. Graph the solution on the number line.



Use the following context to answer questions 5-6.

Franklin is a member of the high school band. Each day he spends an average of 55 minutes practicing for an upcoming band concert. He spent less time practicing on school days, but more time on the weekends. Each of his daily practice times was within 27 minutes of his average.

5. Write an absolute value inequality that represents m, the number of minutes Franklin spends practicing for the band concert on a given day.

   $|m - 55| \leq 27$

6. Solve the inequality.

   $m - 55 \leq 27$ $\quad$ $m - 55 \geq -27$
   $\quad +55 \quad +55$ $\quad$ $+55 \quad +55$

   $m \leq 82$ $\quad$ $m \geq 28$

   $28 \leq m \leq 82$

Use the following context to answer questions 7-10.

Hailey made a personal goal of earning $\$130$ per week at her after-school job. Each week, she has been averaging within $\$23.15$ of her goal.

7. Define the variable.

   Let x represent the money Hailey made.

8. Write an absolute value inequality that represents the context.

   $|x - 130| \leq 23.15$

9. Solve the inequality. Write the answer using set-builder notation.

   $-23.15 \leq x - 130 \leq 23.15$
   $106.85 \leq x \leq 153.15$

   $\{x | 106.85 \leq x \leq 153.15\}$

10. Graph the solution on the number line.